Lactate: from waste product to central metabolite

Steps, biochemistry and current understanding — a four-page synthesis plus FAQ

Christian Puta, apl. Prof. · Sports Medicine and Health Promotion · Friedrich-Schiller-University-Jena | Department for Internal Medicine IV (Gastroenterology, Hepatology and Infectious Diseases), Jena University Hospital, Jena, Germany

Exercise Immunology and Health Lab · immunenotes.github.io

2026-10-29 17:22:59 CET

IN BRIEF

Lactate is formed continuously, with or without enough oxygen, because every glycolytic cell must recycle NAD^+. It is not the cause of muscle acidosis, it is a fuel that mitochondria prefer, the main carrier of carbohydrate carbon between organs, and a hormone-like signal. What blood lactate tells us is not whether exercise is "aerobic" or "anaerobic", but whether glycolytic production and mitochondrial clearance are still in balance. That balance, not $\dot{V}O_2max$, is the core variable for performance, training prescription and metabolic health.

1 · How the old picture formed, and why it fell

The molecule's reputation was built before anyone studied it in a healthy, exercising human. Scheele isolated it from sour milk in 1780; in 1843 Scherer found it in the blood of patients dying of septic and haemorrhagic shock. Between 1918 and 1922 Meyerhof showed in isolated, oxygen-deprived frog muscle that glycogen breakdown was coupled to lactic acid accumulation, and shared the 1922 Nobel Prize with A. V. Hill.[1] From these observations grew the textbook chain: no oxygen → lactic acid → acidosis → fatigue.

Brooks' isotope-tracer work in intact animals and humans from the 1980s onward dismantled every link of that chain. At physiological pH the compound exists almost completely as the lactate anion, so "lactic acid" is a misnomer. It is produced under fully aerobic conditions, taken up by other cells, and oxidised; the *lactate shuttle* (1985) is today regarded as established theory rather than hypothesis.[2]

[1] San Millán (2026a) stresses that these were isolated tissue preparations without circulation, hormones or neighbouring organs — a system that can show overflow, but not redistribution.

[2] Brooks 2018, *Cell Metab* 27:757; Brooks et al. 2022, *Exp Mol Med* 54:1332.

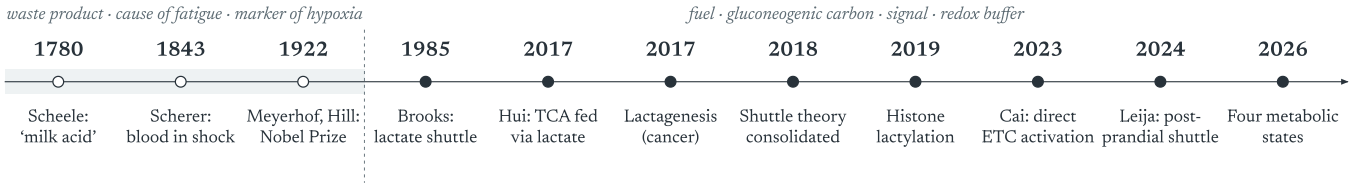


FIGURE 1 Two eras of lactate research (not to scale). Open circles: the observations behind the waste-product model; filled circles: the work that replaced it.

2 · The biochemistry, step by step

The central fact is a redox constraint. Glycolysis reduces NAD^+ to NADH at its sixth step (glyceraldehyde-3-phosphate dehydrogenase). If NADH is not re-oxidised, glycolysis stops. Lactate dehydrogenase (LDH) solves this: reducing pyruvate to lactate regenerates NAD^+ in the cytosol. Because the LDH reaction lies close to equilibrium, every cell running glycolysis makes some lactate, and more glycolytic flux simply means more lactate.[3] Figure 2 follows a single molecule of glucose through a muscle fibre; Table 1 lists the steps.

Overflow follows when glycolytic flux outruns mitochondrial throughput: lactate and protons spill into the blood, the NAD^+/NADH ratio falls, and the falling pH inhibits PFK and calcium release from the sarcoplasmic reticulum, so force declines. San Millán reads this as a built-in governor on runaway glycolysis. Blood lactate is therefore the *net* of production and clearance — which is why the same power can produce 1.5 mmol/L in one cyclist and 7 mmol/L in another with weaker mitochondrial function.[4]

[3] San Millán (2026b): glycolysis does not define intensity; its relation to mitochondrial capacity does.

[4] Example from San Millán (2026a): two cyclists at 300 W. The difference reflects mitochondrial capacity, not effort. Metabolomic profiling of World Tour cyclists: San-Millán et al. 2020, *Front Physiol* 11:578.

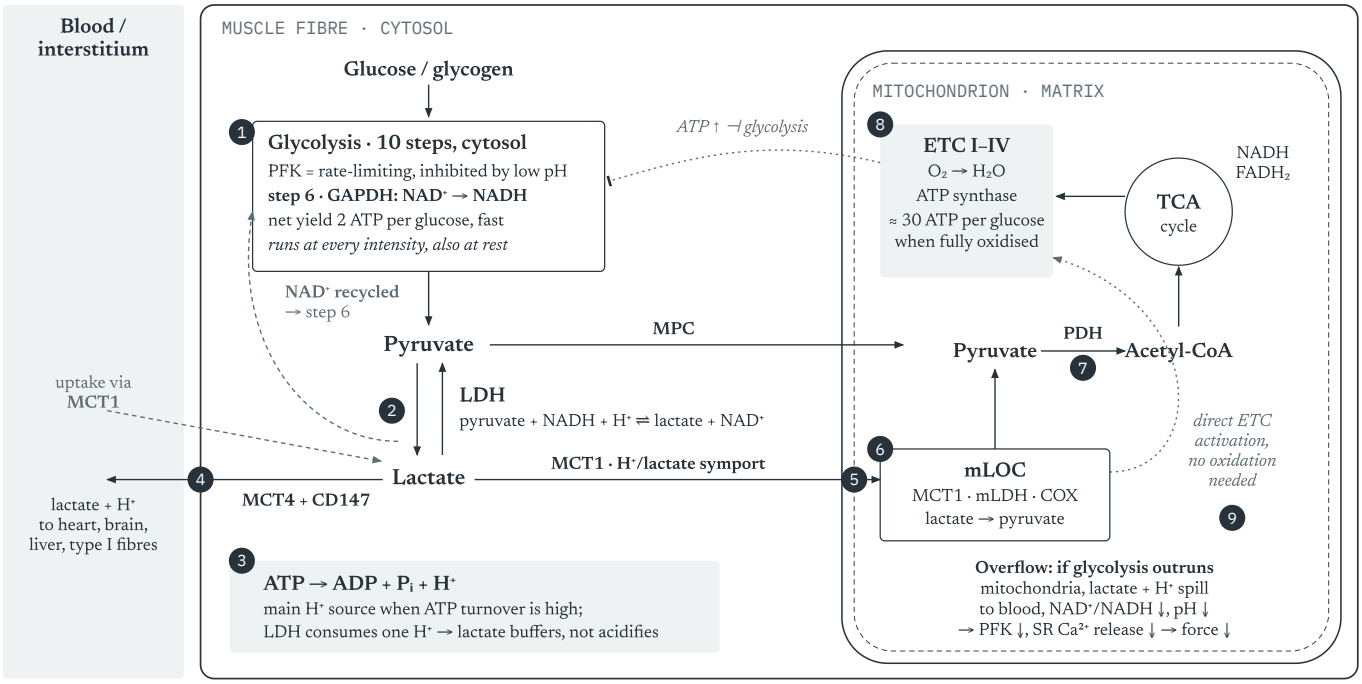


FIGURE 2 Lactate in one muscle fibre. Solid arrows: carbon flow; dashed grey: redox recycling, uptake and signalling; ⊣ inhibition. Numbers refer to Table 1. MPC, mitochondrial pyruvate carrier; MCT, monocarboxylate transporter; mLOC, mitochondrial lactate oxidation complex; mLDH, mitochondrial LDH; COX, cytochrome-c oxidase; PDH, pyruvate dehydrogenase. Own drawing after Brooks (2018, 2022), Cai et al. (2023) and San Millán (2026a, b).

TABLE 1 The sequence from glucose to ATP, read with Figure 2.

#	Step · site	Reaction or carrier	Why it matters
1	**Glycolysis · cytosol**	glucose → 2 pyruvate, net 2 ATP; GAPDH: NAD^+ → NADH	Fast ATP at every intensity; stalls without NAD^+. PFK is the rate-limiting, pH-sensitive enzyme.
2	**LDH · cytosol**	pyruvate + NADH + H^+ ⇌ lactate + NAD^+	Regenerates NAD^+ and consumes one proton: lactate formation sustains glycolysis and buffers.
3	**ATP hydrolysis · myofibrils, pumps**	ATP → ADP + P_i + H^+	The main proton source at high turnover. Lactate is present at the scene but is not the cause of acidosis.
4	**Export · sarcolemma**	MCT4 (with chaperone CD147), H^+-coupled	Glycolytic fibres hand lactate and protons to blood and neighbours.
5	**Uptake · sarcolemma, mitochondria**	MCT1, H^+/lactate symport	Oxidative fibres, heart and brain import lactate; co-transported protons leave the cytosol.
6	**mLOC · inner membrane**	MCT1 + mLDH + COX (+ CD147): lactate → pyruvate	Dedicated machinery to oxidise lactate inside mitochondria — the cell treats it as fuel, not waste.
7	**PDH · matrix**	pyruvate → acetyl-CoA + CO_2 (+ NADH)	Entry point into the TCA cycle for pyruvate from MPC and from mLOC alike.
8	**TCA → ETC · matrix, inner membrane**	NADH, $FADH_2$ → complexes I–IV → O_2; ATP synthase	Full oxidation; O_2 is needed only here, as final electron acceptor.
9	**Signal · matrix**	lactate activates the ETC independent of its oxidation	OXPHOS ↑ and ATP-mediated brake on glycolysis; D-lactate, not metabolised by LDH, works too (Cai 2023).

3 · Three shuttles distribute lactate

Brooks describes three nested systems that move lactate down concentration and redox gradients (Figure 3). Within a cell, lactate formed in the cytosol is oxidised by the mitochondrial reticulum of the same cell. Between cells, glycolytic type II fibres export it through MCT4 and oxidative type I fibres take it up through MCT1. Between organs, the blood carries it from working muscle — and, after meals, from the gut — to heart, brain, liver, kidney and resting muscle, where it is burned or turned back into glucose.

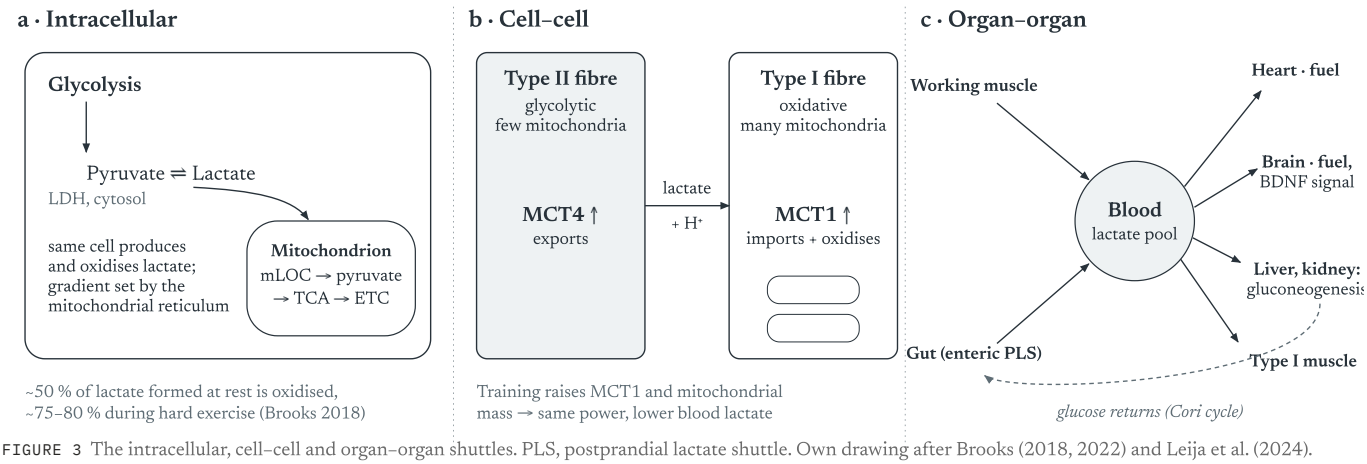


FIGURE 3 The intracellular, cell–cell and organ–organ shuttles. PLS, postprandial lactate shuttle. Own drawing after Brooks (2018, 2022) and Leija et al. (2024).

Three findings give these shuttles their weight (Figure 4). In mice, lactate turns over faster than glucose, and in the fasted state most of glucose's contribution to the TCA cycle arrives indirectly, via circulating lactate, in every tissue except the brain: glycolysis and the TCA cycle are uncoupled at the level of lactate.[5] In humans drinking 75 g glucose, blood lactate rises *before* glucose — the gut epithelium converts part of the load immediately — and a second rise accompanies the glucose peak; about 38 % of the load appears as lactate.[6] And tracer studies show that lactate is mostly oxidised, not stored or excreted.

[5] Hui et al. 2017, *Nature* 551:115. Tumours of lung and pancreas also prefer lactate; pancreatic tumours rely even more on glutamine.

[6] Leija et al. 2024, *Nat Metab* 6:670. Consequence for sports nutrition (Brooks 2023, *Nutrients* 15:2178): the form of carbohydrate matters little, because most passes through lactate.

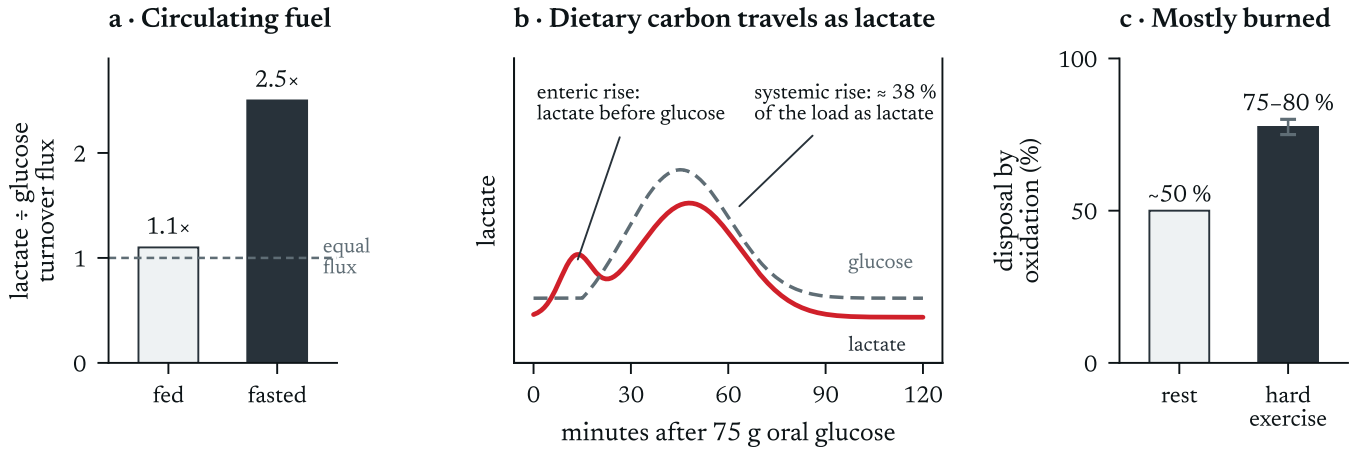


FIGURE 4 Key numbers behind the shuttle concept. a: mice, ^{13}C tracers (Hui 2017); b: humans after a 75 g oral glucose load, curves schematic, timing only (Leija 2024); c: human tracer studies (Brooks 2018).

Lactate as signal: the "lactormone"

Beyond fuel, lactate acts in autocrine, paracrine and endocrine fashion and is now counted among the exerkines.[7] Table 2 sorts the main routes by how directly they have been shown in exercising humans — the column that matters most when reading the newer claims.

[7] Brooks et al. 2022, *Nat Rev Endocrinol* 18:712; 2023, *J Appl Physiol* 134:529.

TABLE 2 Signalling routes of lactate and the strength of the evidence behind each.

Target	Mechanism	Effect	Evidence base
Adipocyte	GPR81 (HCAR1) activation	Lipolysis ↓; fuel use shifts to carbohydrate	Cells, animals (Liu 2009)
Mitochondrial fat import	CPT1 and CPT2 inhibited	Fatty-acid oxidation ↓ at high lactate	Neonatal rat cardiomyocytes (San Millán 2022)
Electron transport chain	Direct activation, metabolism-independent	OXPHOS ↑, glycolysis ↓; T-cell proliferation and effector function ↑	Cell lines, primary T cells (Cai 2023)
Brain	Uptake by astrocytes, neurons	BDNF ↑, hippocampal neurogenesis, executive function	Animals; acute human data (reviews)
Muscle gene programmes	PGC-1α, VEGF, fat-oxidation genes	Mitochondrial biogenesis, angiogenesis	Animal and cell work; reviews
Nucleus	Histone lysine lactylation	Gene expression altered epigenetically	Cells, mice (Zhang 2019)
Tumour	Chronic lactate, LDHA, lactylation	Oncogene and immune-checkpoint expression ↑	Cancer cell lines, partly preprints (San-Millán & Brooks 2017; San-Millán 2020)

4 · Current understanding: one system, four states

San Millán (2026b) replaces the aerobic–anaerobic binary with one metabolic network in four states, defined by whether lactate production and clearance — and with them the NAD^+/NADH ratio — stay matched. Beyond the so-called anaerobic threshold oxidative metabolism still dominates; what is lost is balance, not oxygen. The only truly oxygen-independent phase is the first seconds of a maximal effort, carried by ATP and phosphocreatine — and even that store is refilled oxidatively.[8] Figure 5 places these states over the lactate–performance curve of classical German threshold diagnostics.

[8] Hence the slogan: the sprint is anaerobic in execution but aerobic in regeneration. Repeated-sprint ability depends on mitochondrial capacity.

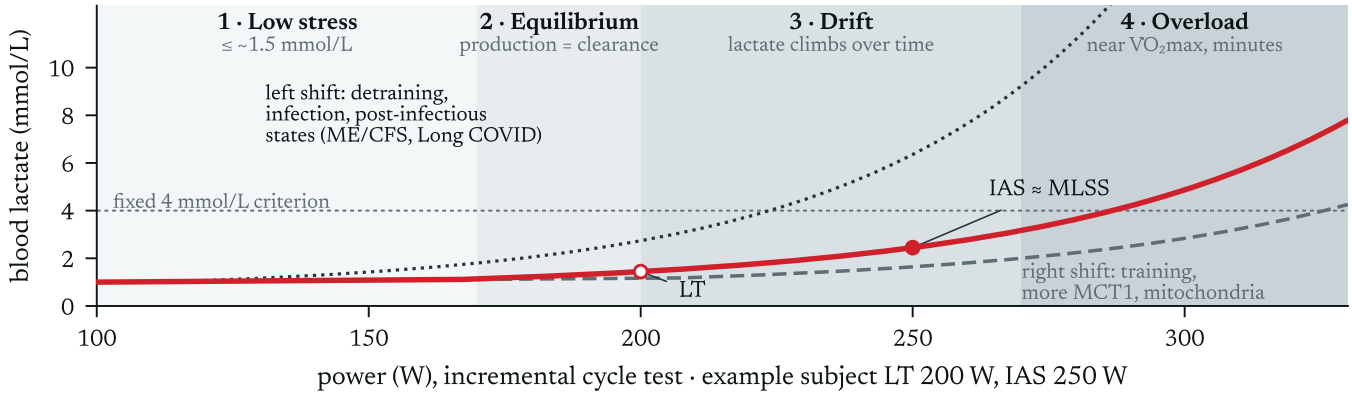


FIGURE 5 Schematic lactate–performance curve with LT and IAS of the lecture's example subject (Kindermann 2004) and the four states (San Millán 2026b). Band limits are approximate; dashed: right shift, dotted: left shift.

TABLE 3 The four states and their nearest equivalents in the two-threshold model.

State	Lactate	Fibres · fuel	Closest classical zone	Sustainable
1 · Low stress	near rest, ≤ ~1.5 mmol/L	type I; fat important	regenerative, below LT	hours
2 · Equilibrium ("zone 2")	formed and cleared at equal rate	type I near full; FATmax	extensive base (GA1) up to ~LT	long
3 · Drift (zones 3–4)	rises progressively	+ type IIa; fat oxidation falls	GA2 to threshold; CP, FTP	min to ~1 h
4 · Overload (zone 5+)	rises fast, pH ↓	+ type IIb; glycolysis maximal, near VO_2max	above IAS; intervals	s to min

Where the sources agree, and where care is needed

Agreed: lactate is fuel, gluconeogenic precursor and signal; blood lactate is the net of production and clearance; training shifts the curve right via MCT1 and mitochondrial capacity. **Terminology:** San Millán's "equilibrium" is stricter than the classical steady state. The IAS is defined as the maximal lactate steady state — elevated but stable, on average near 4 mmol/L — so his "drift" begins below the IAS; zone labels from the two frameworks are not interchangeable.[9] **Evidence level:** TCA-flux data come from mice, ETC activation from cell lines and T cells, lactylation, oncometabolite and checkpoint findings from cell lines (partly preprints) — plausible for the exercising human, not yet shown there. **Two contexts:** exercise lactate is transient and cleared within minutes; tumour lactate is continuous and trapped in an acidic niche. Exercise is not contraindicated in cancer. **Clinical edge:** in ME/CFS and Long COVID, left-shifted curves at low power occur and the IAS may lie below 100 W; zone-based prescription without individual lactate diagnostics can harm. Blood lactate and fat oxidation, measured together, serve as an indirect index of metabolic flexibility (San-Millán & Brooks 2018; San-Millán 2023).

[9] Thresholds also depend on method, step protocol and sampling; fixed 4 mmol/L criteria ignore inter-individual variation (Kindermann 2004; Stegmann 1981).

Literature

Brooks GA (2018) The science and translation of lactate shuttle theory. *Cell Metab* 27:757–785.

Brooks GA et al. (2022) Tracing the lactate shuttle to the mitochondrial reticulum. *Exp Mol Med* 54:1332–1347.

Brooks GA et al. (2022) Lactate as a major myokine and exerkine. *Nat Rev Endocrinol* 18:712.

Brooks GA (2023) What the lactate shuttle means for sports nutrition. *Nutrients* 15:2178.

Cai X et al. (2023) Lactate activates the mitochondrial electron transport chain independently of its metabolism. *Mol Cell* 83:3904–3920.

Hui S et al. (2017) Glucose feeds the TCA cycle via circulating lactate. *Nature* 551:115–118.

Kindermann W (2004) Anaerobe Schwelle. *Dtsch Z Sportmed* 55:161–162.

Leija RG et al. (2024) Enteric and systemic postprandial lactate shuttle phases. *Nat Metab* 6:670–677.

Liu C et al. (2009) Lactate inhibits lipolysis in fat cells through GPR81. *J Biol Chem* 284:2811–2822.

Puta C (2026) ENIF Lecture 6: Lactate — from metabolic waste product to central metabolite. chrisputa.github.io/exercise_nutrition_immune_function

San Millán I (2026a) Lactate: the most misunderstood molecule in biology. Substack, 25 Mar 2026.

San Millán I (2026b) Rethinking exercise metabolism: from "aerobic vs. anaerobic" to metabolic equilibrium. Substack, 10 Apr 2026.

San Millán I et al. (2022) Chronic lactate exposure decreases mitochondrial function… *Front Nutr* 9:809485.

San-Millán I (2023) The key role of mitochondrial function in health and disease. *Antioxidants* 12:782.

San-Millán I, Brooks GA (2017) Reexamining cancer metabolism: lactate production for carcinogenesis could be the purpose and explanation of the Warburg effect. *Carcinogenesis* 38:119–133.

San-Millán I, Brooks GA (2018) Assessment of metabolic flexibility by means of measuring blood lactate, fat, and carbohydrate oxidation responses to exercise. *Sports Med* 48:467–479.

San-Millán I et al. (2020) Is lactate an oncometabolite? *Front Oncol* 9:1536.

San-Millán I et al. (2020) Metabolomics of endurance capacity in World Tour professional cyclists. *Front Physiol* 11:578.

Stegmann H, Kindermann W, Schnabel A (1981) Lactate kinetics and individual anaerobic threshold. *Int J Sports Med* 2:160–165.

Zhang D et al. (2019) Metabolic regulation of gene expression by histone lactylation. *Nature* 574:575–580.

5 · Key messages and FAQ

THE KEY MESSAGES

1. Lactate is always formed, at rest and with ample oxygen — it is the obligatory product of glycolysis and keeps it running by regenerating NAD^+.
2. Lactate is fuel (preferred by heart, brain, type I fibres), a gluconeogenic precursor and a signal; after meals it carries carbohydrate carbon on a scale similar to glucose.
3. Lactate does not cause acidosis; it binds and carries protons and so buffers.
4. Blood lactate is the net of production and clearance and therefore a mirror of mitochondrial capacity.
5. What matters for performance and metabolic health is how long, and up to which intensity, balance is held — not VO_2max alone.

What does lactate do with H^+?

It takes up protons and carries them away. The LDH reaction consumes one H^+ (pyruvate + NADH + H^+ → lactate + NAD^+). The monocarboxylate transporters MCT4 (export) and MCT1 (import) are H^+/lactate symporters: with every lactate anion, one proton leaves the cytosol of the producing cell for the blood, a neighbouring cell or a mitochondrion.

Why is lactate not the cause of acidosis?

The protons come mainly from ATP hydrolysis (ATP → ADP + P_i + H^+) when ATP turnover exceeds mitochondrial resynthesis. Lactate rises at the same time — a correlation, not a cause. Its formation even counteracts acidosis, because it consumes H^+ and secures NAD^+ for glycolysis.[10] A falling pH inhibits PFK and Ca^{2+} release, so fatigue follows from the proton and energy imbalance, not from lactate.

[10] Beyond the four sources: Robergs et al. 2004, *Am J Physiol Regul Integr Comp Physiol* 287:R502. The physicochemical (Stewart) approach instead counts the lactate anion as a strong ion that contributes to pH — the debate is not closed.

What happens to lactate and H^+ in the mitochondria?

Lactate and H^+ reach the mitochondrial lactate oxidation complex together via MCT1 (mLOC: MCT1, CD147, mitochondrial LDH, cytochrome-c oxidase). mLDH oxidises lactate to pyruvate and yields NADH; PDH forms acetyl-CoA for the TCA cycle; NADH and $FADH_2$ feed the electron transport chain. At its end, cytochrome-c oxidase reduces O_2 to water and consumes protons in doing so. In net terms the proton is "burned" together with the lactate — the mitochondria are the real sink for both.

What is lactylation?

A post-translational modification described in 2019: lactyl groups derived from lactate are attached to lysine residues of histones (and other proteins). This changes chromatin structure and so which genes are read — lactate becomes an epigenetic signal as well as a metabolite (Zhang 2019). In tumours, lactylation is linked to proliferation and immune evasion; whether exercise lactate steers adaptation in humans via lactylation is still open.

Why speak of metabolic states rather than a threshold?

Because no switch is flipped. Glycolysis and oxidative phosphorylation run simultaneously at every intensity, and oxygen uptake keeps rising. What changes is the ratio of glycolytic flux to mitochondrial throughput — and with it whether lactate and NAD^+/NADH stay stable, drift slowly or derail. A "threshold" suggests a point; states describe a continuum with transitions. The classical markers (LT, IAS ≈ MLSS) remain useful all the same: they are operational anchors for the borders between states and allow individual training prescription.

What role does lactate play in mitochondrial β-oxidation of fat?

Rising lactate throttles fat oxidation at two levels: via the receptor GPR81 it inhibits lipolysis in adipose tissue, so fewer fatty acids are delivered; in the mitochondria it inhibits CPT1 and CPT2 and thus the import of fatty acids into the matrix, where β-oxidation takes place (Liu 2009; San Millán 2022).[11] Acetyl-CoA from lactate also competes with fat-derived acetyl-CoA for the TCA cycle and the respiratory chain. As a result, fat oxidation peaks in the equilibrium state (FATmax; higher, with lower lactate, in professionals than in less-fit people, San-Millán & Brooks 2018) and falls in drift and overload; training that improves lactate clearance moves this decline to higher power outputs.

[11] Addition: high glycolytic flux also raises acetyl-CoA and malonyl-CoA, which inhibits CPT1 — the classical malonyl-CoA mechanism.

Christian Puta, apl. Prof. · Sports Medicine and Health Promotion · Friedrich-Schiller-University-Jena | Department for Internal Medicine IV (Gastroenterology, Hepatology and Infectious Diseases), Jena University Hospital, Jena, Germany

Exercise Immunology and Health Lab · immunenotes.github.io

Created: 2026-10-29 17:22:59 CET